TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Hydrogen-ready hybrid powertrain retrofit and validation platform for marine propulsion and stationary generator sets			
Korea's 12 National Strategic Technology Fields	Semiconductor ( ), Display ( ), Secondary Battery (√), Biotechnology ( ), Aerospace ( ), Advanced Mobility (√), Next Gen Nuclear ( ), Hydrogen (√), Cybersecurity ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ), AI ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery & Materials		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding & Marine Systems	Main/auxiliary and propulsion system components	Renewable Energy	Hydrogen
Concept of Proposed Technology	This proposal targets a modular engineering and validation platform that enables rapid development and retrofit of low-carbon marine and stationary powertrains. Core elements include: (1) modular digital twin models of medium-speed engines, electric drives, batteries and auxiliaries, (2) supervisory energy and emission management using TNO's Modular Energy Management Strategy (MEMS) software suite, and (3) staged validation using			

	TNO's Innovation Centre for Sustainable Powertrains (ICSP) and Powertrains Test Centre (PTC) in Helmond. Validation assets relevant to this proposal include hydrogen engine test cells (four test cells) for evaluation of combustion and injection concepts, maritime engine dynamometer capability up to 1.4 MW, climatic altitude capability up to 4000 m, and battery testing infrastructure with climate chambers and high-voltage cycling up to 750 V and 400 A (300 kW), with temperature testing from -40 C to +150 C. [R1][R2][R3][R4][R5][R6][R7][R8][R9] The platform is intended to support multiple transition pathways, including hybridization (battery-electric assist and peak-shaving), drop-in e-fuels, and hydrogen internal combustion engine concepts, with a clear route to verification and certification. [R3][R5][R14] The approach builds on published TNO research on modular energy management and model-based co-design (Mentink and Wilkins) and on system-level analyses of electrified heavy-duty and hybrid powertrains (Wilkins, JRC), which are directly transferable to maritime and generator set applications. [R10][R13]
Rationale for Joint R&D Project	Marine vessels and stationary generator sets face increasing pressure to reduce greenhouse gas emissions and local pollutants while maintaining reliability and lifecycle cost. Hybridization and hydrogen-ready engine concepts can reduce fuel consumption and enable future zero- or low-carbon operation, but they introduce significant integration complexity across engine, electric powertrain, battery system and control software. The Korean SME partner contributes application requirements, system integration and field deployment capability. TNO contributes independent model-based engineering methods and unique testing infrastructure for engines (including hydrogen), batteries and full powertrains, enabling faster risk reduction and a stronger evidence base for market introduction and

	certification. [R1][R2][R3][R4][R9]
Key Words	1. Hybrid marine powertrain 2. Hydrogen internal combustion engine 3. Battery energy storage and diagnostics 4. Energy and emission management (MEMS) and digital twin 5. Test-based validation and certification

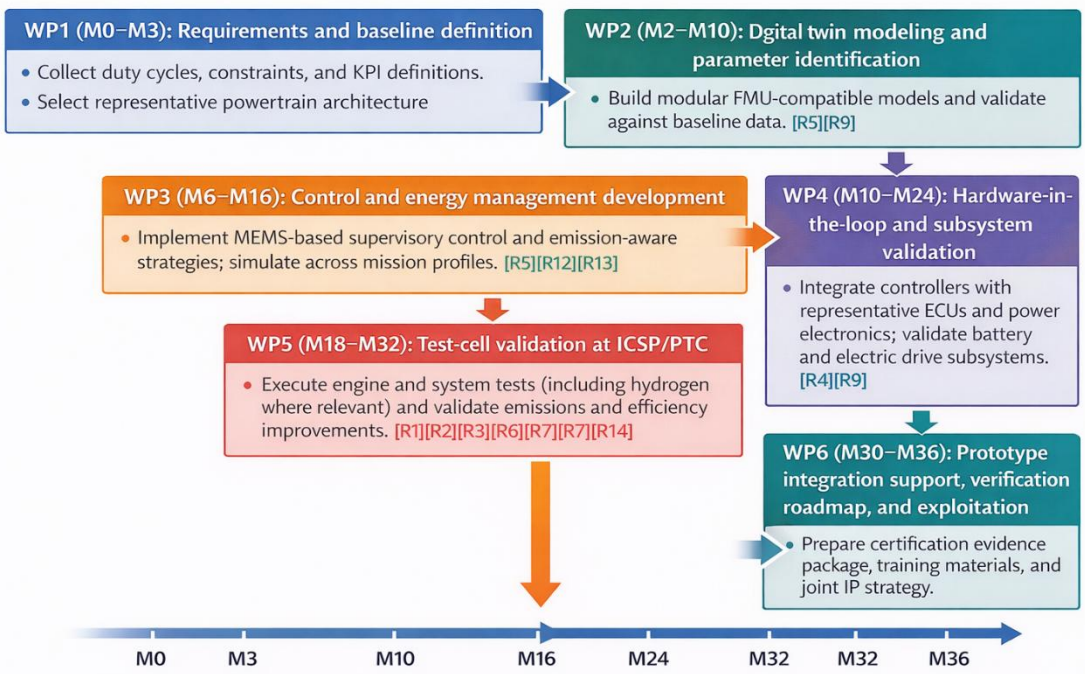
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	• Define reference architectures and duty cycles for hybrid marine propulsion and stationary generator set applications (incl. transient load profiles, operating modes, and safety constraints). • Develop modular digital twin component models (engine, generator, electric drive, battery pack, auxiliaries) and parameter identification workflow, suitable for Software-in-the-Loop and Hardware-in-the-Loop validation. [R5][R9] • Develop supervisory energy management and emission-aware control strategies that can be configured for different fuel pathways (diesel/e-fuels, hybrid, hydrogen ICE), including battery peak-shaving and operating-mode scheduling. [R5] • Validate performance, efficiency and emissions on representative hardware using TNO facilities, including ICSP/PTC test cells and climate-controlled battery testing. [R1][R2][R3][R4][R6][R7][R9] • Deliver a prototype retrofit/control package, test report, and a verification and certification roadmap for market introduction.			
R&D Target Specifications	• Digital twin accuracy (project target): <= 5% RMS error on fuel flow, shaft torque/power, and key thermal states versus measured data over agreed duty cycles. • Control performance (project target): maintain battery state-of-charge within agreed limits (e.g., 30% to 90%) while meeting propulsion or generator load demand; stable transitions between operating modes without power interruptions. • Validation scope: demonstrate staged verification from model-in-the-loop to HIL to test-cell experiments, including engine dynamometer testing (up to 1.4 MW capability), climatic			

altitude capability up to 4000 m, and hydrogen engine operation where applicable. [R3][R6][R7][R8][R9]

- Battery subsystem testing: validate battery pack behavior and safety functions in climate-controlled environments covering sub-zero to high-temperature operation and high-power cycling. [R4][R9]
- Deliverables: documented interfaces and configuration files to enable transfer of models and controllers to the Korean SME integration environment.

R&D Plan



- **WP1 (M0-M3): Requirements and baseline definition.** Collect duty cycles, constraints, and KPI definitions. Select representative powertrain architecture.
- **WP2 (M2-M10): Digital twin modeling and parameter identification.** Build modular FMU-compatible models and validate against baseline data. [R5][R9]
- **WP3 (M6-M16): Control and energy management development.** Implement MEMS-based supervisory control and emission-aware strategies; simulate across mission profiles. [R5][R12][R13]
- **WP4 (M10-M24): Hardware-in-the-loop and subsystem validation.** Integrate controllers with representative ECUs and power electronics; validate battery and electric drive subsystems. [R4][R9]
- **WP5 (M18-M32): Test-cell validation at ICSP/PTC.** Execute engine and system tests (including hydrogen where relevant) and validate emissions and efficiency improvements.

	[R1][R2][R3][R6][R7][R14] • WP6 (M30-M36): Prototype integration support, verification roadmap, and exploitation. Prepare certification evidence package, training materials, and joint IP strategy.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide system-level architecture and requirements decomposition, including safety and verification planning. [R1][R2] • Develop digital twin models and parameter identification workflow; deliver simulation artefacts (FMUs, model documentation). [R5][R9][R13] • Develop energy and emission management control strategies (MEMS-based) and support deployment on representative control hardware. [R5][R13] • Plan and execute experimental validation in TNO facilities (engine, hydrogen, battery, and powertrain testing), including emissions and efficiency measurement. [R1][R2][R3][R4][R6][R7][R9][R14] • Provide technical guidance on certification evidence generation, test reporting, and performance benchmarking.
Expected R&R of Korean SME Partner	• Provide application context (target vessel or generator set classes), baseline system data, and functional requirements, while remaining in public deliverables. • Provide or procure representative hardware (engine, generator, electrical components, battery system) for integration and testing, including interfaces and data access where feasible. • Lead mechanical and electrical integration activities for prototype implementation, including packaging constraints, cooling, and safety engineering. • Support field validation in Korea (or other relevant environments) and collect operational feedback data to refine models and controls. • Lead commercialization strategy, supply chain planning, and aftersales considerations for retrofit or new-build offerings.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• A validated hybrid and hydrogen-ready powertrain architecture and control concept for marine propulsion and stationary generator sets, supported by test data and documented KPIs. [R2][R3][R6][R7] • Reusable digital twin model library and calibration workflow enabling faster engineering iterations and reduced integration risk. [R5][R9] • Demonstrated supervisory energy and emission management functions with transferable interfaces to industrial control platforms. [R5][R13] • A verification and certification evidence package: test plans, test reports, datasets (as permitted), and recommended validation sequences. • Joint dissemination: conference or journal publications, and potential patent filings on integration, control, or diagnostic methods (subject to partner agreements).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems

		6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization